

HW2 47-853 Fall 2026

Note: the hints are on the third page. AI usage is NOT allowed.

Problem 1 (Tightness of greedy) We construct a tight example for the greedy algorithm for monotone submodular maximization under a cardinality constraint. Consider a $k \times k$ grid. We consider the k -tuples where exactly one element is picked from each column, i.e., $\{(i_1, 1), \dots, (i_k, k)\}$; denote by $\mathcal{U}$ the collection of all such k -tuples. We define a submodular function over the size $(2k - 1)$ ground set $N = \{(1, 1), (1, 2), \dots, (1, k), (2, k), \dots, (k, k)\}$. For a subset $S \subseteq N$, define $f(S) := |\{T \in \mathcal{U} : T \cap S \neq \emptyset\}|$.

0. (You don't have to write the answer) Why is f submodular?
1. We want to solve $\max_{S \subseteq N} \{f(S) : |S| \leq k\}$. What's the optimal solution and value?
2. Show that the greedy algorithm can perform as badly as $(1 - 1/e) \cdot f(\text{OPT})$ asymptotically (you are allowed to pick your favorite tie-breaking rule).

Problem 2 (Curvature) Let $f : 2^N \rightarrow \mathbb{R}_{\geq 0}$ be a monotone submodular function where $f(\emptyset) = 0$. The *total curvature* of f is defined as

$$\alpha := \max_{e \in N: f(e) \neq 0} \frac{f(e) - f(e \mid N - e)}{f(e)}.$$

Note that $\alpha \in [0, 1]$, and $\alpha = 0$ if and only if f is linear (modular). Therefore, curvature provides a continuous interpolation between submodular and linear functions.

1. For two arbitrary sets $A = \{a_1, \dots, a_k\}$ and O , denote by $A_i := \{a_1, \dots, a_i\} \forall i \in [k]$. Prove that

$$f(O) \leq \alpha \cdot \sum_{a_i \in A \setminus O} f(a_i \mid A_{i-1}) + \sum_{a_i \in A \cap O} f(a_i \mid A_{i-1}) + \sum_{o \in O \setminus A} f(o \mid A)$$

2. Consider maximizing f subject a single matroid constraint. Let A be the set obtained by the greedy algorithm. Prove that

$$f(A) \geq \frac{1}{1 + \alpha} \cdot f(\text{OPT})$$

Problem 3 (Correlation gap) Let $f : 2^N \rightarrow \mathbb{R}_{\geq 0}$ be a monotone submodular function where $f(\emptyset) = 0$. The concave envelope of f is a function $f^+ : [0, 1]^N \rightarrow \mathbb{R}_{\geq 0}$, defined as

$$f^+(x) := \max \left\{ \sum_{S \subseteq N} \lambda_S f(S) : \sum_S \lambda_S \mathbf{1}_S = x, \sum_S \lambda_S = 1, \lambda \geq 0 \right\}.$$

Let $F : [0, 1]^N \rightarrow \mathbb{R}_{\geq 0}$ be the multilinear extension of f . The *correlation gap* of f is defined as

$$\min_{x \in [0, 1]^N: f^+(x) \neq 0} \frac{F(x)}{f^+(x)}.$$

1. For an arbitrary $S \subseteq N$ and $x \in [0, 1]^N$, prove that

$$\langle \mathbf{1}_S, \nabla F(x) \rangle \geq f(S) - F(x).$$

2. Let $P \subseteq [0, 1]^N$ be an arbitrary solvable polytope. Suppose we want to solve $\max_{x \in P} F(x)$, and denote by x^* an optimal solution. Prove that continuous greedy finds a solution $\bar{x} \in P$ satisfying $F(\bar{x}) \geq (1 - 1/e) \cdot f^+(x^*)$.
3. Prove that the correlation gap of f is at least $1 - 1/e$.

Problem 4 (CRS for knapsack) Given a ground set $N = [n]$, weights $a \in [0, 1]^n$, the independent sets of a knapsack are $\mathcal{I} = \{I \subseteq N : \sum_{i \in I} a_i \leq 1\}$. Consider the following *fixed-order* CRS for the natural LP $P_{\mathcal{I}} := \{x \in [0, 1]^n : \sum_{i=1}^n a_i x_i \leq 1\}$. Let permutation $\sigma : [n] \rightarrow [n]$ be a permutation. For $R \subseteq N$, define $\pi(R)$ as the independent set obtained by greedily picking elements from R in the order σ . Formally, initialize $\pi(R) = \emptyset$. For $i = 1, \dots, n$, add $\sigma(i)$ to $\pi(R)$ if $\sigma(i) \in R$ and $\pi(R) \cup \sigma(i) \in \mathcal{I}$.

1. Consider an instance where $a_1 = \dots = a_{n-1} = \frac{\epsilon}{n-1}$ and $a_n = 1$ for some $0 < \epsilon < 1$. What's the balancedness of the fixed-order CRS for σ being the identity map? (For every $b \in (0, 1]$, determine the largest c such that the CRS is (b, c) -balanced.)
2. The previous example shows we need to be a bit more strategic about the ordering. An item $i \in N$ is big if $a_i > \frac{1}{2}$. Consider an ordering σ where all big items come before all small items. Prove that this ordering gives $(b, 1 - 2b)$ -balanced CRS for every $b \in (0, \frac{1}{2}]$.

Hints:

- Problem 2-1. You may start from writing $f(O \mid A) + f(A) = f(O \cup A) = f(A \mid O) + f(O)$.
- Problem 3-3. Define $P := \{x \in [0, 1]^N : x \leq x^*\}$ and use Problem 3-2.
- Problem 4-2. Analyze the probability of selecting a big item and a small item separately. You may use union bound and Markov's inequality.